310001

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310001

B. Tech. III - Sem. (Main) Exam., (Academic Session 2021- 2022)

Aeronautical Engineering

3AN2 – 01 Advanced Engineering Mathematics – I

AG, AN, CE, CR, ECE, EI & C, ME, MH

Time: 2½ Hours **Maximum Marks: 120**

Min. Passing Marks:

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 6 × 3 marks = 18 marks. Candidates have to answer* ***six*** *questions out of* ***ten***

Part – B: *Analytical/Problem solving questions 3 × 10 marks = 30 marks. Candidates have to answer* ***three*** *questions out of* ***seven****.*

Part – C: *Descriptive/Analytical/Problem Solving questions 3 ×24 marks = 72 marks. Candidates have to answer* ***three*** *questions out of* ***five****.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

PART – A

Q.1 Show that $D = \frac{1}{h}\log\left(\frac{1}{1-\nabla}\right)$, where symbols have their usual meaning.

Q.2 Construct a divided difference table for the following –

x :	1	2	4	7	12
f(x) :	22	30	82	106	216

Q.3 What type of function is used in -

(i) Simpson's $\frac{1}{3}$ rule

(ii) Trapezoidal rule

Q.4 Write down Newton's iterative formula for finding square root of a given number.

Q.5 Define Heaviside's unit function.

Q.6 Find the Laplace transform of $te^{-t}\sin 3t$.

Q.7 What is the relationship between the Laplace transform and the Fourier transform?

Q.8 State the Modulation theorem of complex Fourier transform.

Q.9 Find the Z-transform of $\frac{1}{n!}$.

Q.10 State the final value theorem for Z-transform.

PART – B

Q.1 Prove the following identity –

$$u_0 + \frac{u_1 x}{1!} + \frac{u_2 x^2}{2!} + \frac{u_3 x^3}{3!} + \ldots\ldots\ldots\ldots\ldots\ldots = e^x\left(u_0 + x\Delta u_0 + \frac{x^2}{2!}\Delta^2 u_0 + \frac{x^3}{3!}\Delta^3 u_0 + \cdots\ldots\right)$$

Q.2 Find the Polynomial of the lowest possible degree which assumes the values 3, 12, 15, -12 when x has the values 3, 2, 1, -1 respectively.

Q.3 Find the real root of the equation $x^3 - 2x - 5 = 0$ by Regula – Falsi method.

Q.4 State Convolution theorem for Laplace transform and apply to evaluate -

$$L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}.$$

Q.5 Prove that $\int_0^{\infty} \frac{\cos 6t - \cos 4t}{t} dt = \log_e \left(\frac{2}{3}\right)$.

Q.6 Find the Fourier transform of F(x), if -

$$F(x) = \begin{cases} x^2, & |x| < a \\ 0, & |x| > a \end{cases}$$

Q.7 Find the inverse Z-transform of $\frac{2z}{(z-1)(z^2+1)}$.

PART – C

Q.1 (a) Use Gauss's forward formula to find y_{30} given that $y_{21} = 18.4708$, $y_{25} = 17.8144$, $y_{29} = 17.1070$, $y_{33} = 16.3432$, $y_{37} = 15.5154$

(b) If $F(t) = t^2$, $0 < t < 2$ and $F(t + 2) = F(t)$, then find L {F(t)}.

Q.2 (a) Find the approximate value for the real root of $x\log_{10} x - 1.2 = 0$ correct to five decimal places by Newton-Raphson method.

(b) Find the cubic polynomial which takes the following values –

x :	0	1	2	3
f(x) :	1	2	1	10

hence evaluate f(4).

Q.3 (a) Apply Runge-Kutta method to find an approximate value of y for x = 0.2 in steps of 0.1, if $\frac{dy}{dx} = x + y^2$, given that y = 1, where x = 0.

(b) Use transform method to solve –

$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} + x = e^t$ with $x = 2$, $\frac{dx}{dt} = -1$ at $t = 0$.

Q.4 (a) Use Simpson's $\frac{1}{3}$ rd rule to find $\int_0^{0.6} e^{-x^2}\,dx$ by taking seven ordinates.

(b) Find the Fourier cosine transform of e^{-x^2}.

Q.5 (a) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, (x > 0,\ t > 0)$ subject to the conditions –

(i) $u = 0$, when $x=0,\ t > 0$

(ii) $u = \begin{cases} 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases}$ when $t = 0$

(iii) u (x, t) is bounded

(b) Using the Z-transform, solve $y_{n+2} + 4y_{n+1} + 3y_n = 3^n$ with $y_0 = 0,\ y_1 = 1$.
